

1,550nm MQW-DFB Modulator Integrated Laser

FLD5F10NP-A

FEATURES

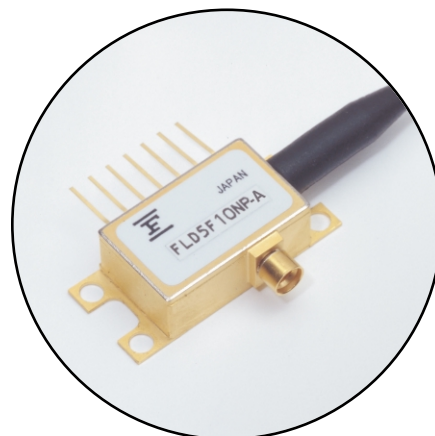
- Modulator Integrated DFB Laser Diode Module
- CW operation of DFB laser section
- Modulation voltage applied only to modulator section
- High speed butterfly package with SMP connection
- Built-in optical isolator, monitor photodiode, thermistor, and thermo-electric cooler
- $\lambda/4$ shifted MQW-DFB chip

APPLICATION

This MI laser is intended for short reach applications (<5km) at 10Gb/s.

DESCRIPTION

The Modulator Integrated DFB Laser (MI DFB Laser) has an electro-absorption modulator monolithically integrated with a conventional Distributed Feed-Back (DFB) $\lambda/4$ shifted Multi QuantumWell (MQW) laser. The modulation voltage is applied to the modulator section while the laser section operates CW allowing extremely low wavelength chirping. Extinction ratios of more than 8.3 dB can be achieved with 2.6 Vp-p modulation. The MI laser is installed in a butterfly type package. The module incorporates a highly stable YAG welded optical coupling system. The module includes an optical isolator, monitor photodiode, thermistor and a thermo-electric cooler.



ABSOLUTE MAXIMUM RATINGS (T_{op}=25°C, unless otherwise specified)

Parameter	Symbol	Condition	Ratings	Unit
Operating Case Temperature	T _{op}	-	-20 to +65	°C
Storage Temperature	T _{stg}	-	-40 to +70	°C
Optical Output Power	P _f	CW	5	mW
Laser Forward Current	I _F	CW	150	mA
Laser Reverse Voltage	V _R	CW	2	V
Modulator Forward Voltage	V _M	CW	-5 to +1	V
Photodiode Forward Current	I _{DF}	-	1	mA
Photodiode Reverse Voltage	V _{DR}	-	10	V
TEC Voltage	V _c	-	2.5	V
TEC Current	I _c	-	1.4	A
Lead Soldering Time	T _{sold}	260°C	10	sec

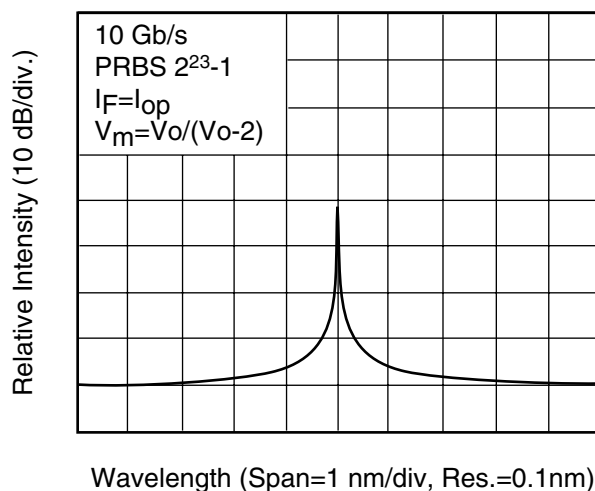
OPTICAL & ELECTRICAL CHARACTERISTICS ($T_L = T_{set}$, $T_C = 25^\circ\text{C}$, BOL, unless otherwise specified)

Parameter	Symbol	Test Condition	Limits			Unit
			Min.	Type	Max.	
On Level Modulation	V_O	-	-1	-0.3	0	V
Modulator Drive Voltage	V_{mod}	$(V_O - V_{mod}) \geq -3.3\text{V}$, $R_{ext} = 8.3\text{dB}$	-	-	2.6	V
Threshold Current	I_{th}	CW, $V_m = V_O$	-	-	30	mA
Operating Current	I_{op}	CW, $V_m = V_O$	50	-	100	mA
Dispersion Penalty	dP	Note (1)	-	-	2	dB
Optical Output Power (Avg. Power)	P_f	Note (1)	-2	-	-	dBm
Forward Voltage	V_F	CW, $I_F = I_{op}$, $V_m = V_O$	-	1.4	2.0	V
Extinction Ratio	R_{ext}	$f = 10\text{Gb/s}$, $I_F = I_{op}$, $V_m = V_O / (V_O - V_{mod})$	8.3	-	-	dB
Peak Wavelength	λ_p	Note (2)	1530	-	1565	nm
Sidemode Suppression Ratio	SSR	Note (2)	35	-	-	dB
Rise Time	T_r	Note (2), 20 to 80%	-	20	40	ps
Fall Time	T_f	Note (2), 20 to 80%	-	20	40	ps
Input Impedance	Z	CW, $V_m < 0$	-	50	-	Ω
RF Return Loss	S_{11}	$f = \text{CD-5GHz}$, 50Ω Test Set, $V_m = V_O$, $I_F = I_{op}$	8	-	-	dB
RF Return Loss	S_{11}	$f = 5\text{-}10\text{GHz}$, 50Ω Test Set, $V_m = V_O$, $I_F = I_{op}$	3	-	-	dB
Cut-off Frequency	S_{21}	-3dB bandwidth, $V_m = V_O - 0.5(V_{mod})$, $I_F = I_{op}$	10	-	-	GHz
Relative Intensity Noise	RIN	$f = 10\text{GHz}$, $V_m = V_O$, $I_F = I_{op}$, $ORL \geq 24\text{dB}$	-	-	-120	dB/Hz
Optical Isolation	I_s	$T_c = -20$ to $+65^\circ\text{C}$	25	35	-	dB

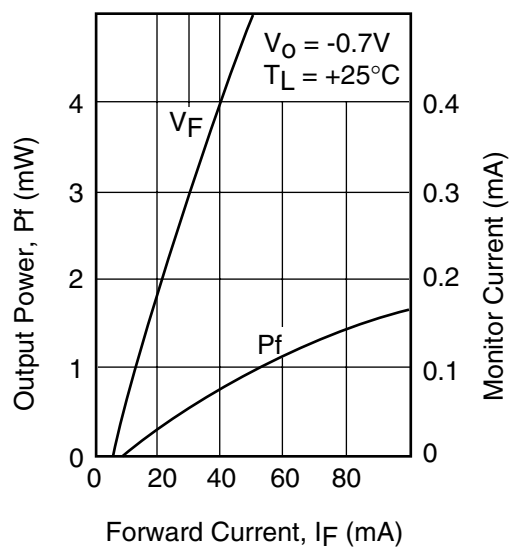
Note (1) FUJITSU Test System
 9.95328Gb/s, PRBS, $2^{23}-1$, $I_F = I_{op}$, $V_m = V_O$ and $(V_O - V_{mod})$
 Dispersion = 82ps/nm, Dispersion penalty at
 Bit Error Rate = $1.0\text{E-}10$

Note (2) FUJITSU Test System
 9.95328Gb/s, PRBS, $2^{23}-1$, $I_F = I_{op}$, $V_m = V_O$ and $(V_O - V_{mod})$

Fig. 1 Lasing Spectrum



**Fig. 2 Forward Current vs. Output Power and
Forward Current vs. Forward Voltage**



**Fig. 3 Extinction Ratio
vs. Modulation Voltage**

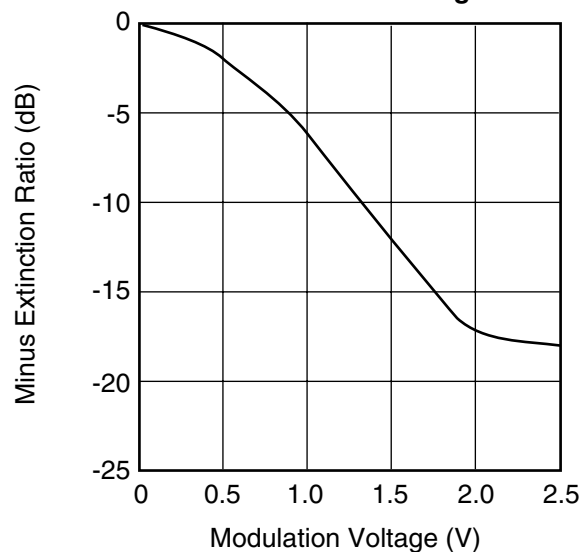


Fig. 4 Cut-off Frequency (S21)

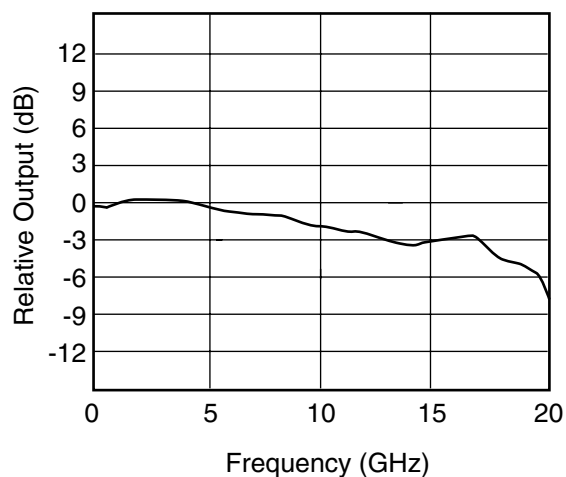


Fig. 5 RF Return Loss (S11)

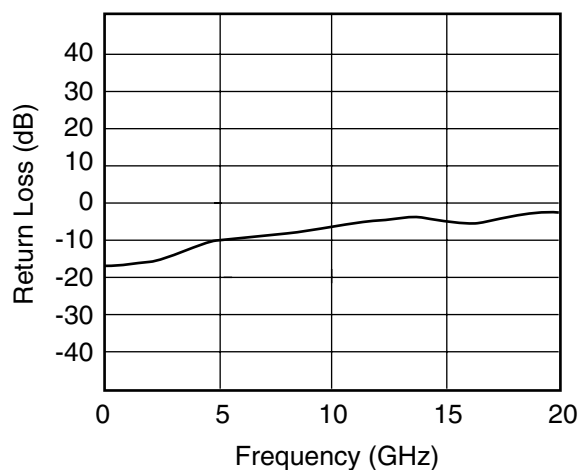
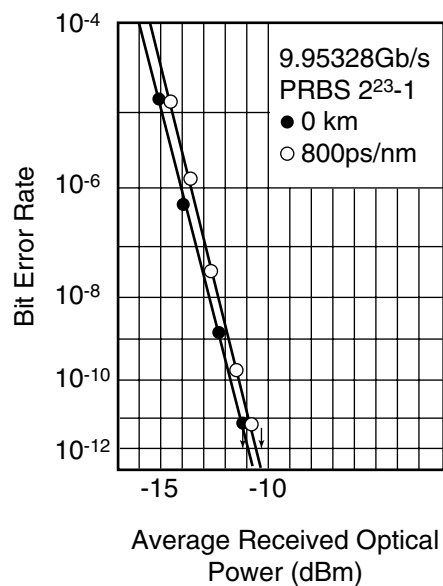
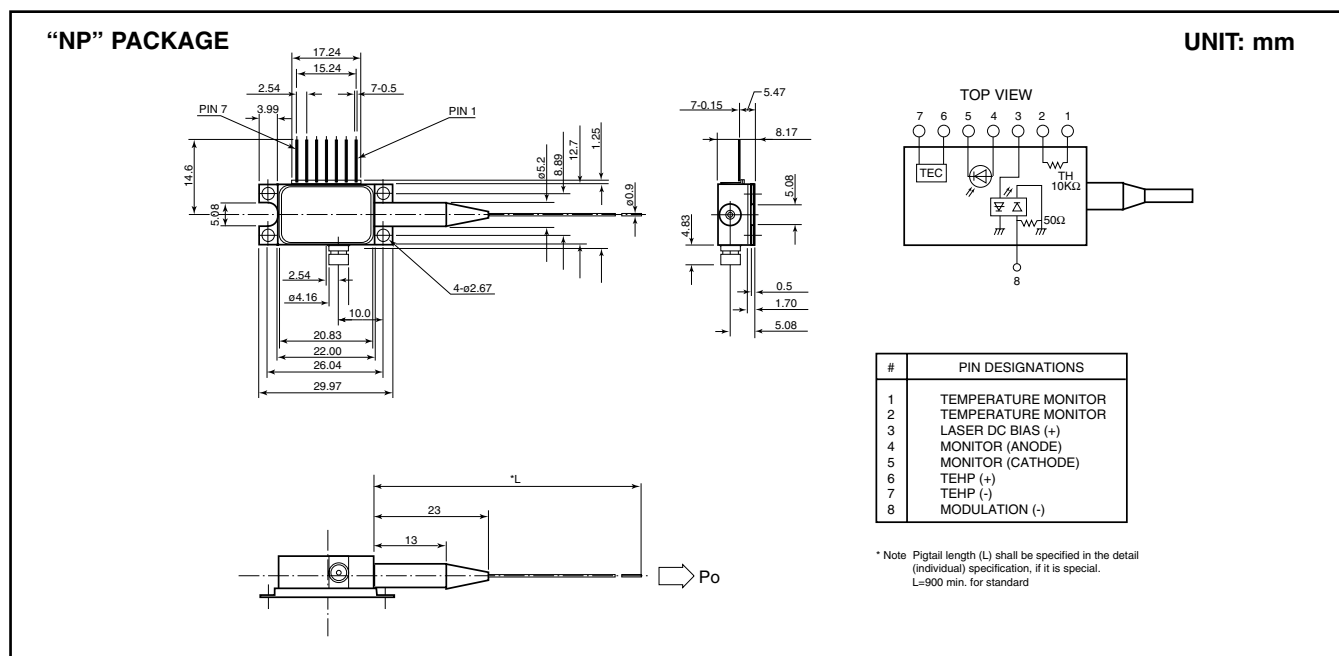


Fig. 6 Transmission Characteristics



1,550nm MQW-DFB _____ **FLD5F10NP-A**
Modulator Integrated Laser

Notes



For further information please contact:

FUJITSU COMPOUND SEMICONDUCTOR, INC. Americas & R.O.W.

2355 Zanker Rd.
San Jose, CA 95131-1138, U.S.A.
Phone: (408) 232-9500
FAX: (408) 428-9111
www.fcsi.fujitsu.com

FME, QDD

Fujitsu Microelectronics Europe GmbH
Quantum Devices Division
Network House
Norreys Drive
Maidenhead, Berkshire SL6 4FJ
United Kingdom
TEL: +44 (0) 1628 504800
FAX: +44 (0) 1628 504888

FUJITSU QUANTUM DEVICES SINGAPORE PTE LTD.

Hong Kong Branch

Rm. 1101, Ocean Centre, 5 Canton Rd. Tsim Sha Tsui,
Kowloon, Hong Kong
TEL: +852-23770226
FAX: +852-23763269

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FUJITSU QUANTUM DEVICES LIMITED

Global Business Division
Global Sales Support Department
Shinjuku Daiichiseimei Building, 2-7-1 Nishishinjuku,
Shinjuku-ku, Tokyo, 163-0721, Japan
TEL: +81-3-5322-3356
FAX: +81-3-5322-3398